Aaron Nguyen

657-626-7999 | aaronngu@usc.edu | linkedin.com/in/aaronnguyen26 | github.com/aaronnguyen26

Education

University of Southern California — Los Angeles, CA
Bachelor of Science in Economics and Data Science — *Aug 2026 – May 2030*

Technical Skills

Languages: Python, Java, C/C++, C#, MATLAB, HTML/CSS
Libraries: PyTorch, XGBoost, scikit-learn, nymPy, pandas, Matplotlib
Tools: Git, GitHub, Jupyter Notebook, VS Code

Projects

Ecoplate AI: Food waste analyzer | *Python, OpenCV, YOLOv8* — July 2025 – Present
- Built a computer vision pipeline using YOLOv8 segmentation and OpenCV to detect leftover food on trays
- Estimated weight, cost, calories, and carbon from a 2.5D volume model and a 45×35 cm tray reference
- Added motion-triggered auto scans, SQLite logging, and a live web dashboard

EPL Football Prediction | *Python, XGBoost, scikit-learn* — Aug 2025 – Nov 2025
- Scraped 8 seasons of FBref data with BeautifulSoup to build a Premier League match dataset
- Engineered 18 features, including Elo ratings, 5-match rolling form, rest days, and xG
- Trained XGBoost models with a Poisson scoreline simulator, reaching 54–58% outcome accuracy

Polymer Property Prediction | *Python, PyTorch, scikit-learn* — July 2025 – August 2025
- Built a Graph Attention Network in PyTorch Geometric to predict 5 polymer properties from RDKit graphs
- Used a multi-task design (3 GATConv layers, 4 heads) with one output head per property
- Trained with the competition's weighted MAE loss, masking missing labels across merged datasets

Experience

AI Research Program Participant — June 2025
Stanford Summer Institute — *Online*
- Designed and trained a custom 3-layer CNN in PyTorch for binary wildfire classification on 2,700+ images
- Preprocessed and split 2,700+ images (128x128, normalized) into 1,887 train / 402 val / 410 test sets
- Benchmarked dropout regularization against a baseline CNN, improving validation accuracy from 80.9% to 81.8%

Research Mentee — Oct 2024 – Apr 2025
Troy University — *Troy, AL*
- Researched quantum interference and superposition models under Professor Trung Hoa Dinh to evaluate Deutsch's algorithm
- Presented findings to 80+ attendees at MathFest '25

IB Computer Science Subject Lead — Jan 2025 – June 2025
Revision Dojo — *Online*
- Co-authored a 10+ page technical pseudocode guide and standardized mark schemes; validated 400+ assessment questions for logical consistency
- Analyzed user interaction data to identify engagement patterns, informing content decisions for a platform serving 300,000+ students

Leadership

Debate Club — Oct 2022 – May 2026
President
- Founded ISHCMC's first debate club in 30 years, grew membership to 35+ students using recruitment surveys
- Designed a comprehensive training curriculum and led 20+ workshops and sparring sessions.

Teens Lead Teens — Dec 2021 – June 2022
Mentor
- Led 20+ hours of bi-monthly personal growth workshops for 100+ peers
- Forged partnerships with SOS Children's Village, DHL Express, and Shinhan Life; initiative recognized in 2 national news publications